

Multimodal Disentanglement: Unsupervised Clustering of Hybrid-Language Music via Convolutional β -Variational Autoencoders

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Abstract

The exponential proliferation of digital music libraries necessitates robust, automated systems for content organization, yet traditional Music Information Retrieval (MIR) approaches often falter when applied to hybrid-language corpora that defy Western-centric taxonomic norms. This research presents a comprehensive investigation into the unsupervised clustering of a hybrid musical dataset comprising English and Bengali (Bangla) tracks, utilizing a deep generative framework. We introduce a multimodal Convolutional β -Variational Autoencoder (β -VAE) designed to fuse spectral audio features with semantic lyric embeddings, thereby bridging the “semantic gap” between acoustic signal and linguistic content. The study integrates the canonical GTZAN genre collection with curated Bengali Automatic Speech Recognition (ASR) and lyric datasets to simulate a complex, cross-cultural information retrieval environment. We evaluate the efficacy of learned latent representations against linear baselines (Principal Component Analysis) and deterministic Autoencoders (AE) using silhouette analysis and density-based metrics. Our empirical results reveal a fundamental trade-off between the regularization required for latent disentanglement and the geometric compactness necessary for traditional clustering algorithms. While the deterministic AE achieved superior cluster separability (Silhouette Score: 0.378), the β -VAE (Silhouette Score: 0.099) offered distinct advantages in interpretability and generative potential, highlighting the “disentanglement-clusterability paradox” in deep unsupervised learning. This report details the methodological pipeline, rigorous experimental protocols, and theoretical implications of applying generative models to the “Benglish” cultural sphere.

1 Introduction

1.1 The Challenge of Hybrid-Language Music Information Retrieval

The digitization of global musical heritage has democratized access to diverse cultural expressions, yet it has simultaneously created a crisis of categorization. Music Information Retrieval (MIR) systems, predominantly trained on Western popular music (WPM) datasets, often struggle to navigate the complex, non-linear feature spaces of non-Western or hybrid musical traditions. This challenge is particularly acute in the South Asian context, where the linguistic landscape is characterized by “code-switching”—the fluid alternation between languages, most notably English and Bengali (Bangla). This phenomenon, colloquially termed “Benglish,” manifests in music not just as a linguistic curiosity but as a distinct socio-cultural signal, blending Western instrumentation with indigenous vocal styles and lyrical themes [3].

Traditional supervised classification methods rely on predefined genre labels (e.g., “Pop,” “Rock,” “Classical”) that are often ill-suited for this hybrid domain. A Bengali “Rock” song may share acoustic properties with Western “Metal” but possess the lyrical sentiment of a traditional “Adhunik” (modern) track. Furthermore, manual annotation of such vast, heterogeneous collections is prohibitively expensive and prone to annotator bias. Consequently, unsupervised clustering emerges as a critical

methodology for discovering latent structures, sub-genres, and cross-cultural similarities without reliance on noisy or nonexistent ground truth labels.

1.2 The Multimodal Imperative

Music is inherently multimodal, conveying meaning through the interplay of acoustic signals (timbre, rhythm, pitch) and symbolic content (lyrics, metadata). Unimodal approaches that analyze audio alone fail to capture the semantic “soul” of a track, while lyric-only models miss the affective intensity of the performance. In hybrid-language scenarios, the semantic gap is widened by linguistic barriers. A purely text-based model might fail to cluster an English “Love Song” with a Bengali “Prem Geeti” (Love Song) if the translation models are imperfect, whereas an acoustic model might group them based on similar tempos and harmonic structures. Conversely, acoustic models might incorrectly separate them based on production quality or instrumentation differences.

To bridge this gap, this research proposes a multimodal deep learning framework that fuses Mel-spectrograms (representing audio) with dense lyric embeddings (representing text). We hypothesize that a joint latent representation, learned via a generative model, can capture the non-linear correlations between these modalities, facilitating clusters that reflect both sonic similarity and semantic coherence.

1.3 The Generative Approach: VAEs vs. AEs

While dimensionality reduction techniques like Principal Component Analysis (PCA) and deterministic Autoencoders (AE) are standard baselines for feature compression, they often lack the distributional smoothness required for generative tasks and meaningful interpolation. The Variational Autoencoder (VAE), introduced by Kingma and Welling (2014) [8], offers a probabilistic alternative by learning a continuous latent space regularized by a prior distribution. By extending this to the β -VAE framework [10], we introduce a hyperparameter β to control the degree of “disentanglement”—the separation of independent generative factors. This study rigorously investigates whether the disentangled representations favored by β -VAEs enhance or hinder the clustering process compared to the unregularized, “broken” manifolds learned by deterministic AEs.

2 Theoretical Background and Related Work

2.1 Benchmarking Genre: The GTZAN Legacy and Limitations

The foundation of automatic music genre classification was laid by Tzanetakis and Cook in 2002 with the release of the GTZAN dataset [11]. Comprising 1,000 thirty-second audio tracks evenly distributed across ten genres (Blues, Classical, Country, Disco, Hip-hop, Jazz, Metal, Pop, Reggae, Rock), GTZAN established the early standards for feature extraction, focusing on timbral texture, rhythmic content, and pitch histograms. For two decades, it has served as the de facto benchmark for validating MIR algorithms [13].

However, the dataset is not without significant flaws. Comprehensive analyses by Sturm (2012, 2013) exposed issues including mislabeling, repetitions (the same track appearing in multiple folds), and audio distortion [15]. Sturm argued that many “state-of-the-art” results on GTZAN were actually fitting to these artifacts—such as identifying a specific radio station’s filter profile—rather than learning musical genre concepts. Despite these warnings, GTZAN remains ubiquitous. In this study, we utilize GTZAN as a representative sample of Western musical structures, while acknowledging its limitations by focusing on unsupervised feature learning rather than supervised accuracy, thereby mitigating the risk of overfitting to incorrect labels.

2.2 The Landscape of Bengali MIR

Research into Bengali Music Information Retrieval has historically been constrained by data scarcity. Early computational efforts focused on stylometric analysis of lyrics, exemplified by the “BanglaMusicStylo” dataset (Hossain & Marouf, 2018), which curated lyrics from legendary figures like

Rabindranath Tagore and Kazi Nazrul Islam for authorship attribution tasks. These studies demonstrated that lyrical features could effectively distinguish between “Rabindra Sangeet” and “Nazrul Geeti,” laying the groundwork for text-based genre analysis.

More recently, researchers have attempted audio-based classification using custom datasets. Bhowmik and Chowdhury (2019) and subsequent studies [19] employed Machine Learning classifiers (SVMs, KNNs) and Deep Neural Networks (DNNs) to classify genres like “Adhunik,” “Band,” and “Folk” from MP3 files. However, these datasets often lacked the scale and standardization of Western benchmarks. The release of large-scale Bengali ASR datasets, such as the “Open Large Bengali ASR Data” by SKNahin [21] and “OOD-Speech” [23], has revolutionized the field by providing thousands of hours of diverse audio, including dialectal variations and varying recording qualities. Our research leverages these contemporary resources to construct a robust Bengali audio corpus.

2.3 Multimodal Fusion Strategies

The integration of audio and lyrics has been shown to significantly enhance MIR tasks. Laurier et al. (2008) demonstrated that multimodal systems outperformed unimodal baselines in mood classification, as lyrics often contain explicit emotional cues that audio features (like tempo) might obscure (e.g., a sad lyric over an uptempo beat). Fusion strategies generally fall into three categories: early fusion (concatenating features before input), late fusion (aggregating model predictions), and hybrid fusion (learning joint representations).

Deep learning has enabled more sophisticated hybrid fusion. Approaches using Deep Boltzmann Machines and cross-modal attention layers [25] allow models to weigh the importance of each modality dynamically. In the generative domain, Vechtomova et al. (2018) utilized VAEs to generate lyrics conditioned on artist-specific audio embeddings, validating the capability of VAEs to model the conditional probability $P(\text{Lyrics}|\text{Audio})$ [26]. Our work adopts a mid-level fusion strategy, concatenating latent audio representations with projected lyric embeddings within the VAE bottleneck.

2.4 Variational Autoencoders and β -Disentanglement

The Variational Autoencoder (VAE) differs from standard autoencoders by imposing a probabilistic constraint on the latent space z . Instead of mapping an input x to a fixed point, the encoder predicts the parameters of a distribution (usually μ and σ of a Gaussian). The model is trained to maximize the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - D_{KL}(q_{\phi}(z|x)||p(z)) \quad (1)$$

The first term encourages reconstruction fidelity, while the second term (Kullback-Leibler divergence) forces the latent distribution towards the prior $p(z) = \mathcal{N}(0, 1)$. Higgins et al. (2017) proposed the β -VAE, introducing a weighting factor β to the KL term [10]. Increasing $\beta > 1$ promotes disentanglement, ensuring that distinct latent dimensions encode independent generative factors (e.g., pitch, timber). However, this comes at the cost of the “information bottleneck,” where high regularization degrades reconstruction quality [28]. This study explores how this regularization impacts the clustering structure of the latent space.

3 Dataset Curation and Preprocessing

A significant contribution of this research is the assembly of a “Hybrid-Language” dataset that mirrors the linguistic reality of modern listening habits. We constructed a composite corpus by aligning high-resource English datasets with lower-resource Bengali collections.

3.1 Dataset Composition

1. **English Audio Corpus (GTZAN):** We sourced the standard genres-original directory from the GTZAN dataset. This subset contains 1,000 audio files in .wav format, sampled at 22,050

Hz. The files are divided into 10 genres with 100 tracks each. This serves as the structural baseline for Western musical forms, providing classes like “Rock,” “Jazz,” and “Metal” that have distinct spectral signatures [11].

2. **Bengali Audio Corpus (ASR & Speech):** The Bengali component was derived from the “Bangla speech/song dataset,” encompassing approximately 2,000 files. This collection relies heavily on the “Open Large Bengali ASR Data” curated by SKNahin and data from the “BanglaBeats” and “OOD-Speech” initiatives [21]. Unlike the studio-quality musical excerpts of GTZAN, this dataset represents “Out-of-Distribution” (OOD) audio, containing a mix of high-fidelity songs, noisy field recordings, and speech/recitation. This variance introduces a realistic challenge for the clustering algorithm: distinguishing between “music” and “speech” as a primary structural dichotomy before identifying sub-genres.

3. Lyric Corpora:

- **English Lyrics:** We utilized the “English Music Lyrics 5 Genres” dataset (d3stron/Anurag Ghosh), which contains over 500,000 samples aligned with genres like Country, Rap, and Rock [31]. We performed stratified sampling to select 800 lyrics that thematically aligned with the GTZAN genres.
 - **Bengali Lyrics:** Lyrics were sourced from the “Bangla Song Lyrics” dataset by Shakirul Hasan Khan [32]. This dataset includes categories like “Rabindra Sangeet,” “Nazrul Geeti,” “Lalon Geeti,” and “Adhunik.” We sampled 800 lyrics to balance the English subset.
4. **Multimodal Alignment:** Since true audio-lyric pairs are often unavailable for scraped datasets, we employed a “synthetic pairing” strategy for the training phase. Audio tracks were aligned with lyric embeddings based on genre labels where available, or randomized within language blocks (e.g., a random English Rock lyric assigned to an English Rock audio track). This approach assumes that genre-consistent lyrics provide a sufficient semantic signal to regularize the audio latent space [8].

3.2 Audio Preprocessing Pipeline

The raw audio underwent a rigorous transformation pipeline to generate input features compatible with the Convolutional encoder.

- **Resampling:** All audio files, regardless of origin, were resampled to a uniform 16,000 Hz (16 kHz). This downsampling (from GTZAN’s 22.05 kHz) was necessary to standardize the input size across the diverse Bengali and English sources.
- **Feature Extraction (Spectrograms vs. MFCCs):** The project explored two representations. The primary analysis in the draft report focuses on Mel-Spectrograms with 64 Mel bands, calculated using librosa. These retain more spectral detail than cepstral coefficients. However, the accompanying code implementation utilized Mel-Frequency Cepstral Coefficients (MFCCs) with 13 coefficients ($N_{MFCC} = 13$) for certain experimental runs [8]. This report primarily discusses the Spectrogram results as they provide a richer feature set for the CNN, but acknowledges the MFCC baseline as a computationally efficient alternative that decorrelates the input, aiding diagonal-covariance VAEs.
- **Temporal Standardization:** To create fixed-size tensors, the spectral frames were either padded with zeros or truncated to a fixed length $T = 130$ (corresponding to approximately 3 seconds of audio). The final input shape for the network was $(B, 1, 64, 130)$ for spectrograms or $(B, 1, 13, 130)$ for MFCCs [8].
- **Normalization:** Input features were log-scaled (to decibels) and min-max normalized to the range $[0, 1]$. This scaling is critical for the Sigmoid output activation of the decoder, ensuring the reconstruction loss is well-defined.

3.3 Text Preprocessing and Embedding

Raw text requires vectorization to be fused with audio latents. We utilized Sentence-BERT (SBERT), a modification of the BERT network that uses siamese structures to derive semantically meaningful sentence embeddings [33].

- **Model:** We selected paraphrase-MiniLM-L6-v2 [34]. This model is distilled for speed and performance, mapping sentences to a 384-dimensional dense vector space.
- **Processing:** Lyrics were truncated to fit the transformer’s context window. The model outputs a fixed-size vector for each lyric document.
- **Rationale:** Unlike “Bag-of-Words” or TF-IDF, SBERT captures semantic context. In a hybrid environment, this is crucial. Ideally, a multilingual model (like `distilbert-base-multilingual-cased`) would align Bengali and English semantics in a shared space. However, utilizing an English-centric model like MiniLM implies that the Bengali lyrics must be either transliterated/translated or that the model relies on the language-agnostic structural properties of the embeddings. The report notes the use of separate CSVs; the effectiveness of MiniLM on raw Bengali text is a noted limitation, effectively acting as a language identifier due to tokenization differences.

4 Methodological Framework

The core of this study is the Convolutional β -Variational Autoencoder, a generative model designed to learn a joint distribution over the multimodal input space.

4.1 The Convolutional Encoder

The encoder network functions as a non-linear feature extractor, compressing the high-dimensional spectrograms into a compact latent vector. We implemented a deep Convolutional Neural Network (CNN) architecture:

1. **Input Layer:** Accepts tensors of shape $(1, 64, 130)$.
2. **Conv Block 1:** Conv2d $(1 \rightarrow 32)$ filters, kernel = 3, stride = 2, padding=1 + BatchNorm + ReLU. This layer captures low-level spectral edges and temporal onsets.
3. **Conv Block 2:** Conv2d $(32 \rightarrow 64)$ filters, kernel = 3, stride = 2, padding=1 + BatchNorm + ReLU. Captures mid-level textures (timbre).
4. **Conv Block 3:** Conv2d $(64 \rightarrow 128)$ filters, kernel = 3, stride = 2, padding=1 + BatchNorm + ReLU. Encodes high-level musical structure.
5. **Flattening:** The volumetric feature maps are flattened into a dense vector h_{audio} (dimension 128) [8].

4.2 Multimodal Fusion Mechanism

To integrate the lyrical content, we employed a late-fusion concatenation strategy within the bottleneck.

1. **Lyric Projection:** The 384-dimensional SBERT embeddings are too large relative to the audio features. We passed them through a Multi-Layer Perceptron (MLP) projection layer: Dense $(384 \rightarrow 64)$. This compresses the semantic information into a 64-dimensional vector h_{text} .
2. **Concatenation:** The audio vector h_{audio} (128-d) and lyric vector h_{text} (64-d) are concatenated to form the fused representation h_{fused} (192-d).

$$h_{\text{fused}} = \text{Concat}(h_{\text{audio}}, h_{\text{text}}) \quad (2)$$

The authors identify this as a “simple fusion” baseline, acknowledging that more complex mechanisms like cross-modal attention could provide better alignment [8]. However, concatenation forces the VAE to utilize information from both modalities to minimize the KL divergence, effectively “gluing” the audio and text manifolds together.

4.3 Latent Parameterization and the β -Objective

The fused vector parameterizes the variational distribution $q_{\phi}(z|x, t)$:

- **Mean Head:** $\mu = \text{Dense}(h_{\text{fused}} \rightarrow 64)$
- **Variance Head:** $\log \sigma^2 = \text{Dense}(h_{\text{fused}} \rightarrow 64)$

The latent variable z is sampled using the reparameterization trick: $z = \mu + \varepsilon \odot \exp(0.5 \cdot \log \sigma^2)$, where $\varepsilon \sim \mathcal{N}(0, 1)$.

The model is trained to minimize the β -VAE loss function:

$$\mathcal{L} = \mathcal{L}_{\text{Recon}} + \beta \cdot D_{KL}(\mathcal{N}(\mu, \sigma^2) || \mathcal{N}(0, 1)) \quad (3)$$

- **Reconstruction Loss ($\mathcal{L}_{\text{Recon}}$):** Mean Squared Error (MSE) between the input spectrogram and the decoder output.
- **Regularization ($\beta \cdot D_{KL}$):** The term that enforces the Gaussian prior. We experimented with $\beta \in [2.0, 5.0]$, with the reported results using $\beta = 3.0$. This relatively high β prioritizes disentanglement—ensuring the latent dimensions are independent—over reconstruction quality.

4.4 The Decoder

The decoder attempts to reconstruct the audio spectrogram from the latent z (dimension 64). Note that the model is not trained to reconstruct the lyrics; the text serves as a conditioning signal for the audio latent space.

1. **Unflatten:** Projects z back to the spatial dimensions of the encoder’s last feature map.
2. **Transposed Convolutions:** A stack of ConvTranspose2d layers (mirroring the encoder) upsamples the feature maps.
3. **Output:** A final layer maps to 1 channel with a Sigmoid activation, producing the reconstructed spectrogram $\hat{x}$.

5 Experimental Setup

5.1 Baselines

To validate the VAE’s performance, we implemented three comparative baselines:

1. **PCA + KMeans:** A linear baseline where Principal Component Analysis reduces the flattened spectrograms to 32-64 components, followed by K-Means. This tests whether the data manifold is simple enough for linear methods.
2. **Deterministic Autoencoder (AE) + KMeans:** An identical neural architecture to the VAE but without the variational sampling layer and without the KL divergence loss. This isolates the effect of the probabilistic prior on clustering.
3. **Spectral Features + KMeans:** Direct clustering on raw spectral features (MFCC means/variances), a traditional MIR approach.

5.2 Training Protocol

- **Environment:** The models were trained on a GPU-accelerated environment (simulated Kaggle/Colab).
- **Optimizer:** Adam optimizer with a learning rate of 10^{-3} .
- **Batch Size:** 32 (for stability) or 64.
- **Epochs:** 15 epochs for the VAE, 12 epochs for the AE (sufficient for convergence of the reconstruction loss) [8].
- **Data Split:** The dataset of 1,600 samples was used for unsupervised training. No test set separation was strictly necessary for clustering, but validation splits were used to monitor reconstruction loss.

5.3 Clustering Algorithms and Metrics

Post-training, we extracted the latent representations (using the mean vector μ for the VAE). For the final clustering stage, we further enriched these features by concatenating one-hot encoded genre hints where available, creating a vector $F = [z|\text{genre}]$. This vector was normalized (StandardScaler) before clustering. We applied three algorithms:

1. **K-Means:** With $k = 8$, chosen to approximate the natural stratification of the data (Bangla Speech, Bangla Song, Western Pop, Rock, etc.).
2. **Agglomerative Clustering:** Hierarchical clustering with 8 clusters.
3. **DBSCAN:** Density-based clustering, requiring tuning of ϵ and min_samples .

Evaluation Metrics: Since unsupervised tasks lack absolute ground truth, we relied on internal validation metrics:

- **Silhouette Score:** Measures cluster cohesion and separation (Range: -1 to +1; higher is better).
- **Davies-Bouldin Index (DBI):** Ratio of within-cluster scatter to between-cluster separation (Lower is better).
- **Adjusted Rand Index (ARI):** Measures agreement with external genre labels (Range: -1 to +1).

6 Results and Analysis

6.1 Quantitative Clustering Performance

The following table summarizes the performance of the multimodal VAE against the baselines.

Table 1: Comparative Clustering Results

Method	Silhouette Score ($\uparrow$)	DBI ($\downarrow$)	ARI
VAE + KMeans	0.099	2.385	-0.0004
PCA + KMeans	0.129	2.089	-0.0007
AE + KMeans	0.378	0.813	-0.0017

Analysis of the Deterministic Advantage: The deterministic Autoencoder (AE) drastically outperformed the VAE and PCA in terms of cluster definition, achieving a Silhouette Score of 0.378 compared to the VAE’s 0.099. This result, while initially counter-intuitive given the VAE’s theoretical sophistication, highlights a critical dynamic in unsupervised learning. The AE, unconstrained by

the KL divergence term, is free to fracture the latent space. It can map distinct data modes (e.g., “noisy Bengali speech” vs. “clean English pop”) to widely separated islands in the latent manifold to maximize reconstruction accuracy. This discontinuity creates “easy” boundaries for K-Means to find [9].

Analysis of the VAE’s Performance: The β -VAE performed poorly on clustering metrics (Silhouette 0.099, DBI 2.385). This is a direct consequence of the $\beta = 3.0$ regularization. The KL term forces the posterior distributions to overlap and conform to the standard normal prior $\mathcal{N}(0, 1)$. This “smoothing” effect fills the gaps between clusters, creating a continuous, globular manifold ideal for interpolation and generation but detrimental for separation. The high β essentially “melted” the clusters together, making the latent space too dense for K-Means to identify distinct centroids.

Analysis of ARI: The ARI scores were negligible across all methods (≈ 0). This confirms that the clusters found by the models (based on acoustic/semantic similarity) did not align with the noisy genre labels provided. For example, the model likely grouped “male speech” (from the Bengali dataset) and “rap vocals” (from English lyrics) together based on timbre, defying the “Genre” labels. This suggests that in hybrid datasets, “Genre” is a poor proxy for the underlying structural reality.

6.2 Latent Space Visualization

Visual inspection of the t-SNE projections corroborated the quantitative metrics:

- **AE Latents:** Exhibited sharp, distinct clusters pushed to the edges of the space.
- **VAE Latents:** Showed a centralized “cloud” structure. While there was a gradient separating Bengali and English samples (likely due to the “Speech vs. Song” dichotomy), there were no clear margins. This confirms that the Gaussian prior successfully regularized the space, preventing the formation of the “holes” that define the AE’s clusters.

6.3 The Failure of Density-Based Clustering

Initial experiments with DBSCAN on the fused VAE latents resulted in all points being classified as noise (-1). This is a manifestation of the “curse of dimensionality” combined with VAE regularization. Because the VAE compresses all data points into a high-density region around the origin (the “Gaussian ball”), the density differences required by DBSCAN become indistinguishable without extremely precise hyperparameter tuning.

7 Discussion

7.1 The Disentanglement-Clusterability Paradox

This study illuminates a fundamental tension in deep unsupervised learning: the objective of disentanglement opposes the objective of clustering.

- **Generative Disentanglement:** Requires a continuous, smooth latent space where moving along one axis changes one factor (e.g., tempo) continuously. This implies no gaps.
- **Clustering:** Requires a discontinuous, fragmented space with clear gaps between groups.

By selecting $\beta = 3.0$, we prioritized the generative capability of the model. The resulting representations are likely more interpretable (dimensions might correlate with meaningful factors) but less separable. Future applications prioritizing retrieval over generation should prefer deterministic AEs or lower β values ($\beta < 1$).

7.2 Challenges in Multimodal Fusion

The “simple fusion” (concatenation) strategy employed revealed limitations. Concatenating a 128-d audio vector with a 64-d lyric vector assumes a static relationship between the modalities. However, in “Benglish” music, the semantic load shifts. A song might have English verses (high semantic value for the MiniLM model) and a Bengali chorus (low semantic value/noise). The concatenation treats these equally, potentially introducing noise into the latent space. The results suggest that the audio modality (spectrograms) likely dominated the clustering, as the text embeddings for untranslated Bengali lyrics would cluster essentially as “unknown language” artifacts.

7.3 Code-Switching as a Latent Factor

The clustering results imply that the model primarily learned to distinguish production modes rather than cultural genres. The most distinct separation observed was likely between “Speech” (from the Bengali ASR dataset) and “Music” (from GTZAN). This suggests that unsupervised clustering in hybrid-language contexts is effective for Audio Event Detection (Music vs. Speech) but requires more refined, language-aligned text embeddings (e.g., using Multilingual BERT) to capture the subtleties of “Benglish” genre fusion.

8 Conclusion and Future Directions

This research successfully established a reproducible pipeline for the unsupervised clustering of hybrid-language music, integrating standard Western benchmarks with novel Bengali resources. The implementation of the Convolutional β -VAE demonstrated the capability to fuse Mel-spectrograms and lyric embeddings into a joint latent space.

Key Findings:

1. **Deterministic Superiority:** For pure clustering tasks, unregularized Autoencoders provide significantly better cluster separation (Silhouette 0.378) than VAEs (Silhouette 0.099).
2. **Regularization Trade-off:** High β values promote disentanglement at the expense of cluster definition, creating an over-smoothed manifold.
3. **Infrastructure for Hybrid MIR:** The integration of the SKNahin ASR dataset and d3stron’s lyric corpus proves to be a viable strategy for low-resource language analysis.

Recommendations: To resolve the disentanglement-clusterability paradox, future work should explore Vector Quantized-VAEs (VQ-VAE) [35]. VQ-VAEs enforce a discrete latent space, which naturally aligns with the categorical nature of clustering while retaining the generative properties of VAEs. Additionally, implementing Contrastive Learning (SimCLR) [37] could explicitly optimize the latent space for clustering by pulling similar multimodal pairs together, providing a more robust objective than simple reconstruction. Finally, employing Multilingual BERT (mBERT) [33] is essential to truly bridge the semantic gap in “Benglish” lyrics.

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